The Geometry of Learning: How One Gradient Step Alters Latent Manifold Structure

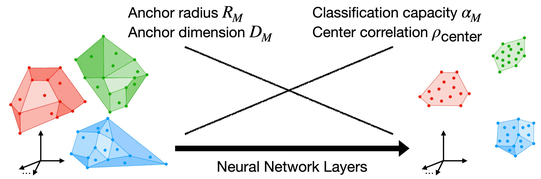
Shaunak Bhandarkar*[1,2], Chi-Ning Chou*[2], SueYeon Chung[2,3]

Affiliation 1: Stanford University, Stanford, CA 94305 *Affiliation 2:* Center for Computational Neuroscience, Flatiron Institute, New York, NY 10010

Affiliation 3: Center for Neural Science, NYU, New York, NY 10003

Motivations

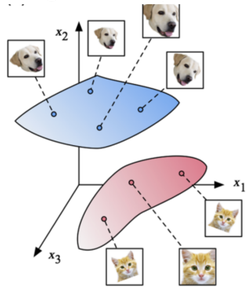
A large endeavor in theoretical neuroscience is to discover the learning rules and and neural architectures present across the brain. Prior empirical work in artificial neural networks has involved inferring plausible learning rules from a set of candidates based on certain measured observables [2, 5].



Using Manifold Capacity Theory [3], a framework that uses statistical physics to link linear classification capacity of manifolds to their intrinsic geometric features, we aim to develop a theory of how these observables (e.g., manifold radius, dimension) for neural representations change over *a single application* of a learning rule. This could enable us to infer learning rules from their high-level geometric "signatures".

What is a Neural Manifold?

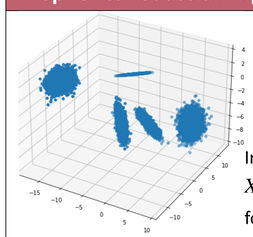
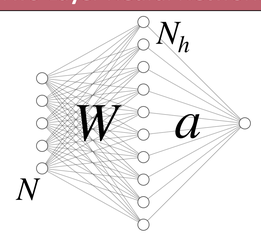
Neural manifolds arise as the locus of the different possible activity states of a population of neurons in response to the same object. A natural goal for biological and artificial networks is to classify manifolds. Geometrically, we can understand neural manifolds by their *radius* and *dimension*; such quantities have proven relevant in analyzing biological mechanisms, such as processing of auditory information in parietal cortex [6].



Example depictions of manifolds for 'dog' and 'cat' (Fig 1(b) from [3]).

Research Question: Can we empirically and theoretically characterize how one *gradient descent step* in an artificial neural network causes manifold representational geometry in the hidden layer to change?

Experimental Setup: Spherical Gaussian Manifolds

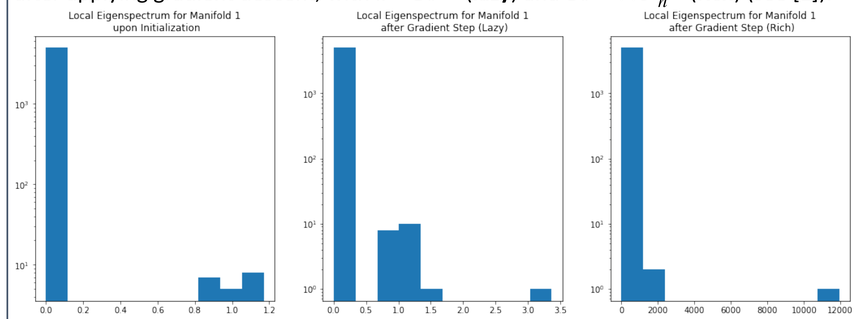
Spherical Gaussian Input Data	Two-Layer Neural Network
 Input data: $X_\mu \in \mathbb{R}^{m\times N}$, for $\mu = 1,\ldots,M$	
Initialize M random d-dimensional spherical Gaussian manifolds of radius R_M inside $\mathbb{R}^N$, and sample m points from each manifold. Each manifold has a class label (1, 2, ..., M).	We feed the input data into a 2-layer network $f_\theta(x) = \frac{1}{\sqrt{N}} a^T \sigma(Wx)$, where σ denotes the tanh nonlinearity. We compare the latent representations for each manifold before and after doing one gradient step.

Linking Eigenvalues to Manifold Geometry

For data sampled from a spherical Gaussian manifold μ, $C_\mu = \frac{1}{m} WX_\mu \left(WX_\mu\right)^T$ or $\frac{1}{m}\sigma\left(WX_\mu\right)\left(\sigma\left(WX_\mu\right)\right)^T$ we can utilize the eigenspectrum of its covariance matrix in determining the manifold radius and dimension.

$$R_\mu = \left(\frac{\sum_{i=1}^{N_h} \lambda_{\mu i}^4}{\sum_{i=1}^{N_h} \lambda_{\mu i}^2}\right)^{1/2} \qquad D_\mu = \frac{\left(\sum_{i=1}^{N_h} \lambda_{\mu i}^2\right)^2}{\sum_{i=1}^{N_h} \lambda_{\mu i}^4}$$

We compare the eigenvalue distributions for the latent covariance matrix before and after applying gradient descent, with lr = 10^{-3} (**lazy**) and $10^{-3} \times N_h^{1/2}$ (**rich**) (see [1]).

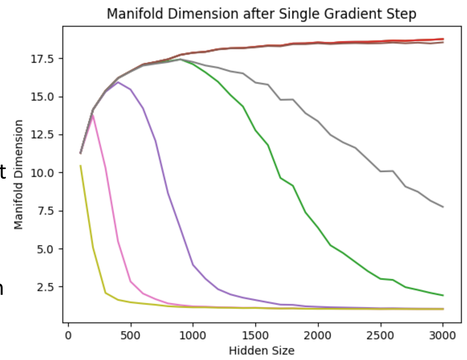


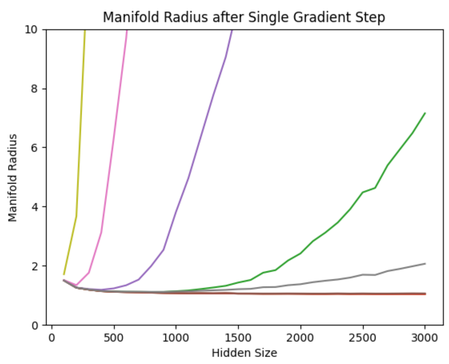
Distributions show the eigenvalue distributions for the latent states prior to applying the tanh nonlinearity. Lazy learning shifts eigenvalue distribution; rich learning effectively does a rank-1 update.

(Parameters used: M = 10, m = 3000, d = 20, N = 400, R_M = 1, N_h = 5000)

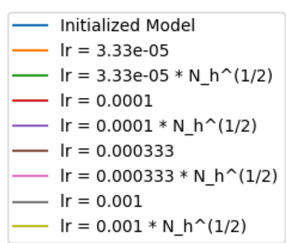
Empirical Analysis for One Gradient Step

We use the radius and dimension formulas for the spherical Gaussian manifold case to empirically compute these quantities for latent manifolds (pre-nonlinearity) both before and after applying a single gradient descent step. Here we show our empirically simulated results (averaged over 3 independent simulations) across different network hidden sizes for both the lazy and rich learning regimes:





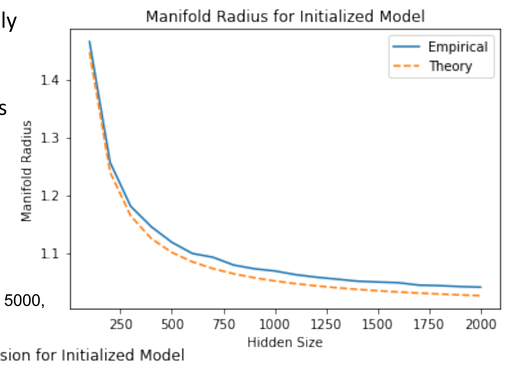
(Parameters used: M = 10, m = 2000, d = 20, N = 100, R_M = 1)



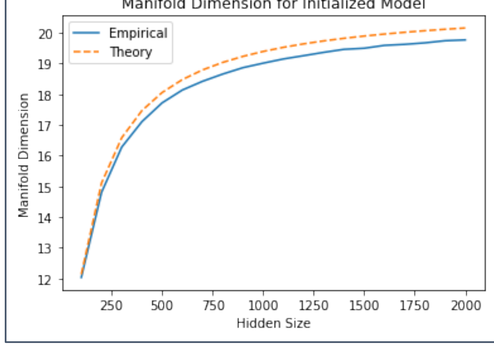
Observation: A gradient update in the rich regime appear to play a larger role in dimension reduction whereas a single update in the lazy regime seems to control radius from growing.

Theoretical Calculation of Radius and Dimension

We wish to theoretically compute radius and dimension asymptotically in terms of the hidden size N_h. For the untrained 2-layer neural network, this is achieved via random matrix theory.



(Parameters used: M = 10, m = 5000, d = 21, N = 100, R_M = 1)



Using random matrix theory, we hope to understand how a single gradient step shifts the existing eigenvalue distribution. This will enable us to understand how radius and dimension change!

Summary & Future Directions

- Future steps involve investigating phenomena arising in wide neural network limit ($N_h \to \infty$):
  - Is there a double-descent phenomenon present for manifold radius in the rich learning regime? How does manifold dimension converge?
- Computations in the neocortex are well-modeled by artificial neural networks (e.g. visual processing in the ventral visual stream <—> CNNs)
  - We hope to apply our theory to detect the presence (or absence) of gradient-like computations

Main References

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Acknowledgements

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